

3. Non-Preemptive Priority Scheduling algorithm

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Non-Preemptive Priority Scheduling algorithm

Completed

Write a C program to implement the Non-Preemptive Priority Scheduling algorithm. The program should accept number of processes their Arrival Time (AT), Burst Time (BT), and Priority for each process. The process with the highest priority (lowest numerical value) must be executed first among those that have arrived. Calculate the Average Waiting Time and Average Turnaround Time.

Input Format

The first line contains an integer representing the number of processes.

The next lines contain three integers for each process, where

- The first integer denotes the Arrival Time (AT)
- The second integer denotes the Burst Time (BT)
- The third integer denotes the Priority of the process.

A lower numerical value indicates a higher priority.

Output Format

The output displays the Average Turnaround Time and the Average Waiting Time.

Example 1

Sample Input 1

```
3
0 0 2
1 3 1
2 2 3
```

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, completed = 0, current_time = 0;
5
6     printf("Enter number of processes: ");
7     scanf("%d", &n);
8
9     int at[n], bt[n], pr[n];
10    int ct[n], tat[n], wt[n], visited[n];
11
12    for(i = 0; i < n; i++) {
13        printf("\nEnter AT, BT and Priority for P%d: ", i + 1);
14        scanf("%d %d %d", &at[i], &bt[i], &pr[i]);
15        visited[i] = 0;
16    }
17
18    while(completed < n) {
19        int min_index = -1;
20        int highest_priority = 1e9;
21
22        for(i = 0; i < n; i++) {
23            if(at[i] <= current_time && visited[i] == 0) {
```

Test Cases 0

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2

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First-Come,First-Served CPU Scheduling Algorithm

Completed

Write a C program to implement the First-Come, First-Served (FCFS) CPU Scheduling Algorithm. The program should take the number of processes, their Arrival Times (AT), and Burst Times (BT) as input. Calculate and display the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process, along with the Average Turnaround Time and Average Waiting Time.

To evaluate the performance of the algorithm, we use the following formulas:

- **Completion Time (CT):** The time at which a process finishes execution.
- **Turnaround Time (TAT):** Total time taken from arrival to completion.
- $TAT = CT - AT$
- **Waiting Time (WT):** The time a process spends waiting in the ready queue.
- $WT = TAT - BT$

Input Format

1. The first input line contains an integer representing the number of processes.

Select Language: C (GCC 9.2.0)

```
26 float total_tat = 0, total_wt = 0;
27
28 for (int i = 0; i < n; i++) {
29     tat[i] = ct[i] - at[i];
30     wt[i] = tat[i] - bt[i];
31
32     total_tat += tat[i];
33     total_wt += wt[i];
34 }
35
36 float avg_tat = total_tat / n;
37 float avg_wt = total_wt / n;
38
39 printf("\nAverage Turnaround Time: %.2f", avg_tat);
40 printf("\nAverage Waiting Time: %.2f\n", avg_wt);
41
42 return 0;
43 }
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1